TAIYANG CHEN

+1 617-238-4079 | taiyang_chen@hms.harvard.edu

EDUCATION

Harvard University, Harvard Medical School
Master of Science in Biomedical Informatics — Sept 2023 – expected Dec 2024

- **GPA: 4.00/4.00**
- Selected courses: Mathematics in Biology, Computing Skills for Biomedical Informatics, Computational Statistics for Biomedical Science, Biomedical Artificial Intelligence, Deep Learning for Biomedical Data, Advanced Topics in Data Science

Southern University of Science and Technology (SUSTech) — Shenzhen, China
Bachelor of Science in Biology — Sept 2019 – Jun 2023

- Major in Bioinformatics, **GPA: 3.85/4.00, ranking 1st**
- Selected courses: Computational Biology, Bioinformatics, Biostatistics, Introduction to Computer Programming (Java), Artificial Intelligence, Data Structure and Algorithm Analysis, Probability and Statistics, Genomics, Cell Biology

WORKING PAPERS

- **T. Chen**, S. Hou, H. Wang and Y. Peng. "Machine learning based single-cell analysis of acute myeloid leukemia data". International Conference on Bioinformatics and Intelligent Computing. 2024 (under review)
- N. Xiong, **T. Chen**, L. McBride, et al. "Leveraging machine learning to resolve biological confounding and expedite lymphoma diagnosis". American Society of Human Genetics (ASHG). 2024 (meeting abstract submitted)

RESEARCH EXPERIENCES

Machine Learning for Cancer Detection and Classification Using Liquid Biopsy Data — **Mar 2024 – present**
Dana-Farber Cancer Institute
advisor: Dr. Mark A. Murakami

- Processed mutation, clonality and infectious diseases data from liquid biopsy, which contains around 200 genetic variants in 853 lymphoma patients, 36,467 non-lymphoma cancer patients and 60 normal samples
- Trained machine learning models to identify genetic variants that might be causal to cancer; used random forest and neural network to model non-linear relationship and interaction between predictors
- Compared model performance; found random forest showed 0.98 AUC in cancer detection
- Interpreted model outputs using coefficients and feature importance; used local interpretable model-agnostic explanations to explain each individual prediction in test data

Machine learning based analysis of single-cell acute myeloid leukemia data — **Jul 2022 – Aug 2022**
Massachusetts Institute of Technology
advisor: Prof. Manolis Kellis

- Preprocessed single-cell gene expression data (gene * cell) containing about 20,000 genes from bone marrow of three individuals spanning four months
- Built a machine learning model based on recurrent network structure to perform dimensionality reduction of single-cell data and identify the major cell types; improved the model by applying deep embedded clustering and modifying model layers; observed clear separation between different cell types
- Applied the model to study acute myeloid leukemia (AML) at the single-cell level; identified the cell types in the data and changes in cell type composition by time

- Compared cell type composition and conducted differential expression analysis in samples from AML patients and healthy donors; revealed ribosomal genes are upregulated in the AML condition

Dimensionality Reduction and Visualization of Single-cell RNA Sequencing Data **Apr 2021 – Jun 2023**
Southern University of Science and Technology
advisor: Prof. Zhen Zhang and Prof. Wenfei Jin

- Developed a neural network based on stacked autoencoders that could preserve both the local and the global structure of single-cell gene expression data after dimensionality reduction and clustering
- Used layer-wise pretraining, self-supervised fine-tuning, deep embedded clustering and consistency constraints to improve model performance
- Tested the model on eight different scRNA-seq datasets from software simulation, human and mouse

Immune Responses of stimulated human and mouse T Cells to SARS-CoV-2 virus **Sept 2020 – Jun 2023**
Southern University of Science and Technology
advisor: Prof. Xi Chen and Prof. Wenfei Jin

- Stimulated T cells separately with three different surface antigens of SARS-CoV-2 virus and one surface antigen of adenovirus
- Compared flow cytometry data from SARS-CoV-2 stimulated T cells with the unstimulated and adenovirus-simulated T cells; concluded that low T cell maturity might reduce mouse immune response
- Performed quality control, dimensionality reduction, clustering, marker gene identification, cell type annotation, differential expression analysis and cell composition comparison analysis on single-cell RNA-seq data of human and mouse T cells using the Seurat package in R
- Identified up-regulated interferon stimulated genes, indicating the importance of the interferon antiviral pathway in human and mouse immune response

AWARDS AND HONORS

- Outstanding Graduate from the Department of Biology, SUSTech 2023
- Outstanding Thesis of the Department of Biology, SUSTech 2023
- Third-prize Scholarship, SUSTech 2022
- Third-prize Scholarship, SUSTech 2021
- Scientific research and innovation project fund, SUSTech (10,000 CNY) 2021
- Second-prize Scholarship, SUSTech 2020
- Third prize, Undergraduate Life Sciences Contest, Guangdong 2020
- Exemplary College Ambassador Team of Winter Holiday Activity, SUSTech 2020
- Freshmen Scholarship, SUSTech 2019

SKILLS

- Computational languages: R, Python, Java, Matlab, Linux, C++, SQL
- Machine learning: dimensionality reduction, classification, regression, clustering